



ELECT MAINT Department
SIMHADRI

Note No :1



Ref:SMTTP/ELECT MAINT/2024-25/EMDLMI9/654761

Date:

08/11/2024(11:08:20)

Subject:LMI for OVER FLUXING OF GENERATOR TRANSFORMER
LMIOINELEC009

LMIs are reviewed for every two years for further improvement and incorporation of recommendation by COS if any. COS had reviewed the LMI for over fluxing of generator transformer in May-24 and no changes were recommended.

As a part of periodical review of station LMI's, LMI/ OIN/ELEC/009 for "OVER FLUXING OF GENERATOR TRANSFORMERS" is due for review in NOV-24. No changes envisaged.

Submitted for review & approval of competent authority please.

Submitted by:

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Note No:4

It is requested to modify as per standard format of Simhadri LMI (Guidelines for Issue of LMIs as a Controlled Document) LMI/OGN/GEN/002 Rev:04 as uploaded on Simhadri EEMG LAN.Checklist annexure should be last part of LMI

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Note No:5

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Note No:6

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Note No:7

LMI is in line with OIN.

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Note No:8

Reviewed and submitted for approval

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Note No:9

May please be approved.

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On:26/11/2024(20:54:15) **Ref:**SMTTP/ELECT MAINT/2024-25/EMDLMI9/654761

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

OVER FLUXING OF GENERATOR TRANSFORMER

LMI/ OIN/ELEC/009 Rev:1

NOV - 2024

APPROVED FOR IMPLEMENTATION BY HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI Name	Over fluxing of Generator transformer
2	LMI No	LMI/ OIN/ELEC/009,REV:1
3	Reference Document	COS-OIN/ELEC/009,REV:1
4	Superseded Document	NIL
5	Prepared by	K.J.Prasanna Kumar Sr.Manager (EM)
6	Checked by	Satish Kumar Singh DGM (EM)
7	Checked by	A Bala Balaji DGM (EM)
8	Checked by	Tenneti Vijay Krishna HOD(EM)
9	Reviewed by	Om Prakash AGM(Maint)
10	Reviewed by	Prasanta Kumar Jena GM(O&M)
11	Approved by	Sameer Sharma HOP (Simhadri)

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TITLE	OVER FUXING OF GENERATOR TRANSFORMER
LMI NO	LMI/OIN/ELEC/009, Rev:1

1. INTRODUCTION:

Generator transformers can be damaged by operating them at flux levels significantly higher than their design value (Modern GTs have flux densities of 1.7 tesla). This can cause overheating and undesirable vibration not only in the core and winding but also in metal parts affected by stray flux. The increased magnetizing current may be sufficient to cause the differential protection to operate.

2. SCOPE:

- (i) This LMI covers Generator transformer of Unit # 1, 2, 3 & 4.
- (ii) Since generator automatic voltage regulator at Simhadri is of Digital (DAVR) nature, and bypassing facility of over fluxing limiter not directly accessible from front panel, there is no possibility of limiter going out of service so long as DAVR is healthy.
- (iii) Similar to above, since there is no bypassing switch available on the panel for GT over fluxing protection relay the said protection will remain in service unless the relay is taken out.

3. REFERENCE DOCUMENTS:

This is based on Operation Instruction Note COS/OIN/OPS/ELEC/009.

4. SUPERSEDED DOCUMENTS: NIL

5. DETAILS OF REVISION: NIL

6. CIRCUMSTANCES IN WHICH OVERFLUXING CAN OCCUR

6.1 On Load

The generator transformer will be over fluxed if a high voltage is applied to either the HV or LV winding. This may result from a high system voltage being impressed on the transformer or when generating leading MVARs. It is unusual for this to occur under normal operating conditions. The transformers most at risk are those with the higher nominal flux

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densities. When generating maximum leading MVARs, it is imperative that the terminal volts of machines operating with higher flux density generator transformers should not be allowed to rise above nominal. If this precaution is observed the generator will reach the AVR VAR limiter setting before over fluxing of the transformer occurs.

6.2 Running at Less than Synchronous Speed

If the AVR is in service during running up or down of a generator or whilst running at less than normal speed with the generator circuit breaker open, depending on the type of AVR fitted, it will attempt to maintain normal machine terminal volts. As the flux in the transformer varies inversely with frequency, the over fluxing will occur to some degree at speeds lower than normal. However, the more modern AVR's are fitted with (Voltage/Frequency) limiters which maintain the (Voltage/Frequency) ratio at nominal while turbo generators are at lower speed and the generator circuit breaker is open.

6.3 Running on No Load with Terminal Voltage Higher than Normal

This can occur during tests, for example, when checking the generator Open Circuit Characteristics or Commissioning an Automatic Voltage Regulator. If the generator transformer is connected, the generator terminal voltage must not be allowed to exceed 105% of rated voltage. With the generator excitation on auto mode and AVR in service, the generator voltage should be regulated within the limits of +5% and –15% of rated voltage. However, when in manual mode of operation (no control of AVR), care is required to ensure that the limit of 105% is not exceeded. Long term danger to transformer can occur when regularly synchronising a unit with an unstable AVR, where frequent small over voltages are applied to the transformer. The resulting excessive magnetostriction between adjacent core plates gradually wears away the core plate insulation at high spots, leading to circulating currents within the core and subsequent core melt down at normal terminal voltage. There is

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more chance of transformer failure through this type of over fluxing than by a large once only incident.

6.4 Sudden Disconnection of Load

The tripping of a generator circuit breaker normally trips the generator field switch, and so the problem of over fluxing may not arise. However, if load throw off is due to system tripping and without opening of generator circuit breaker, the voltage will be controlled within the limits by AVR. If the excitation is on manual the control can only be exercised by the manual, intervention. In such cases the generator voltage may rise beyond rated limits resulting in an over fluxing condition. Therefore, operation of excitation in manual mode i.e. without AVR is not recommended.

6.5 Operation of AVRs

After the replacement of a stator or any other high voltage equipment, phasing out' is required to be carried out before synchronizing the Unit with the grid. During this 'phasing out' test, the dead grid busbar (765kV/400kV/220kV) is charged from the generator through the Unit Transformer. This test should be carried out with AVR and limiter in service to avoid over fluxing condition of Generator Transformer. The over fluxing protection should also be in service to trip the field in case of any abnormality in the AVR or limiter.

Unless the unit (Voltage/Frequency) ratio control limiter is in service, when running up a unit the excitation should not be applied or the AVR switched into service until the Unit speed is between 95% to 100% of rated speed. Normally this feature is in-built as a permissive for field breaker closing.

Over fluxing protection of GT in generator relay protection panel to be co-ordinated with V/f limiter of AVR in Excitation room so that in case of over fluxing, V/f limiter of AVR to act first instead of over fluxing protection which will trip the machine.

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7. PRECAUTIONS AGAINST EXCESSIVE OVERFLUXING:

7.1 Over fluxing Protection:

All 220 kV/400 kV/765 kV connected units have AVRs and over fluxing protection. There are two basic types. The simpler of the two has an over fluxing relay which has the ability to force a reduction in excitation and provide protection if this proves unsuccessful. The second type consists of a control circuit built into the AVR which will keep the voltage/ frequency ratio to within a pre-set value. In the event that these circuits are unable to control a rise in the machine terminal volts, an over fluxing relay is fitted to provide the necessary protection.

It is sometimes considered undesirable for over fluxing protection to be in service when the unit is synchronized, as occurrence of over fluxing is very unlikely when generator is connected to the grid. The protection is therefore inoperative when the line generator breaker is closed. Alternative opinion considers that during conditions of house load and islanded operation, closing the generator breaker without the support of the grid necessitates a requirement for the over fluxing relay.

8. SPECIFIC INSTRUCTIONS TO BE FOLLOWED:

Table-1

S.No	INSTRUCTION	RESPONSIBILITY	DOCUMENT REF.
1	Do not regulate generator terminal voltage beyond 105% of nominal value (21 KV)	Operation	DDCMIS trends

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2	Check auto start set point of DAVR corresponds to 95% of rated voltage and ensure all excitation limiters are in service with recommended setting during unit overhaul.	Electrical Maint. Area in-charge	File:Excitation test results.
3	GT Overfluxing protection relay to be tested during overhauling along with alarm / trip.	Electrical Maint. Area in-charge.	File: Relay test result.
4	If over fluxing limiter or GT over fluxing protection relay is taken out of service for any reason, clearance is to be obtained from Engg. and same is to be recorded in Operation shift Log book.	Electrical Maint. Area in-charge	Shift Log Book

9. METHODOLOGY OF IMPLEMENTATION:

- (i) Sl.no 1 of above table-1 is to be complied by Operation SCE.
- (ii) Sl.no: 2 to 4 of above table-1 are to be complied by Electrical maintenance concerned groups and records are to be maintained.

10. FREQUENCY OF CHECKING AND MEASUREMENTS:

Once in during unit Overhaul.

11. RESPONSIBILITY CENTRE FOR IMPLEMENTATION:

Area in-charge for excitation system and Area in-charge for Testing lab are responsible. Overall responsibility lies with DGM (EM)

12. EXCEPTION REPORTING & DISPENSATION OF DEVIATION:

Any deviation shall be reported to HOD (EM) and can be dispensed with his approval.

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13. RECORD KEEPING:

1. Relays test results.
2. Excitation system testing results.

The above records will be maintained by concerned area maintenance groups.

14. REVIEW :

HOD(EMD) will review & take the approval of competent authority (HOP). The document will be reviewed on 2 yearly basis or earlier, if required.

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15. CHECKLIST FOR OVERFLUXING OF GT

S No.	Description	Responsibility	Remarks
1.	AVR operation in Manual mode is not allowed, whenever AVR comes in manual mode should be immediately informed to EMD	SCE	
2.	Synchronizing the Generator with trouble in AVR is not allowed. No alarm should come from AVR during synchronization	SCE	
3.	While doing the OCC test if generator transformer is connected, the terminal voltage should never be allowed to exceed 5% of nominal value.	AGM(EMD)	
4.	In manual mode of AVR, Generator Terminal Voltage shouldn't exceed 105% of rated	UCE	
5.	While the running up to/ or running down from synchronous speed it will be the responsibility of the Shift Charge Engineer to ensure that the generator terminal voltage is limited to 105% of nominal voltage.	SCE	
6.	If the unit trips on over fluxing the same shall be reported to EMD for further investigation	SCE	
7.	Over fluxing Protection to be tested during overhauling and record to be maintained.	DGM (EMD)	
8.	Co-ordination of AVR Over flux limiter and overflux protection to be checked and recorded during unit OH	DGM (EMD)	